

GRAFT-VERSUS-HOST DISEASE

AT A GLANCE

Acute cutaneous graft-versus-host reaction (GVHR) develops in 20 percent to 80 percent of patients undergoing allogeneic bone marrow transplantation, usually 10 to 30 days after infusion.

Skin is the earliest and most frequent target. Liver and gastrointestinal involvement are also common; elevated total bilirubin levels and diarrhea can aid in establishing the diagnosis.

Predisposing factors include mismatch in major histocompatibility complex (MHC) antigens, transplantation between unrelated individuals, older age, and certain prophylactic regimens.

The eruptions of acute GVHR typically begin as faint, erythematous macules on any skin surface, sometimes with a perifollicular pattern in early stages, and may progress to erythroderma (later stages) or bullae (stage 4).

Pathologic features of acute GVHR include a dermal lymphocytic infiltrate, basal layer vacuolar alteration, and epidermal cell necrosis at a frequency of at least four apoptotic bodies per linear millimeter of epidermis (grade 2).

Chronic GVHR generally occurs more than 60 days after allogeneic marrow transplantation, with an incidence of approximately 25 percent. It is classified into lichenoid and sclerodermoid forms.

Cutaneous eruptions resembling acute allogeneic GVHR can occur after autologous and syngeneic transplantation in up to 10 percent of patients.

The concept of lymphocyte recovery eruption is relevant in stem cell transplantation, as GVHRs often develop during peripheral lymphocyte

recovery and reflect immunologic differences between donor and recipient or result from immunologic manipulation. Some histologically non-specific cutaneous eruptions in patients at risk for GVHD may represent lymphocyte recovery eruptions rather than true GVHR, since acute cutaneous GVHR requires cytotoxic targeting of the epidermis by donor-derived effector lymphocytes.

EPIDEMIOLOGY AND INCIDENCE

Marrow-ablative regimens vary and commonly include combinations of cyclophosphamide, busulfan, total-body irradiation, methotrexate, and thiotepa.

Acute cutaneous GVHR occurs in 20% to 80% of patients, typically 10 to 30 days after stem cell infusion. Risk factors include MHC antigen mismatch, unrelated donor transplantation, older age, and specific prophylactic regimens.

Patients often receive high-dose cyclosporine at the time of transplantation to promote engraftment and reduce GVHD severity. Other strategies to prevent GVHD involve antibody-mediated depletion of putative effector T cells from the graft.

Chronic GVHD is divided into lichenoid and sclerodermoid subtypes. The lichenoid form usually precedes the sclerodermoid form, and both typically manifest beyond 60 days post-transplant. Some patients experience both types sequentially or simultaneously. Prior acute GVHD is the strongest predictor for developing chronic GVHD. The overall incidence of chronic cutaneous GVHD is about 25%. Rare cases have been reported following autologous transplantation.

Cutaneous eruptions mimicking acute allogeneic GVHR occur in up to 10% of patients after autologous or syngeneic transplants. Prospective studies on eruptions in peripheral blood stem cell transplant recipients are limited, but

mild skin eruptions are observed coinciding with peripheral lymphocyte recovery.